CXCR2 expression and vitamin D in early pregnancy

HSE 103 · First project check-in

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Research question

How does maternal whole-blood **CXCR2 expression** vary with baseline vitamin D concentration and subsequent preeclampsia status? This exploratory project focuses on immune signaling. CXCR2 encodes a chemokine receptor involved in immune-cell signaling; its biological role motivates the choice, rather than a genome-wide search for significant results [1].

The continuous covariate is baseline 25-hydroxyvitamin D (ng/mL). The two categorical covariates are pregnancy outcome (normal pregnancy or preeclampsia) and vitamin D category (<30 or 30 ng/mL), derived from the continuous measurement. The cutoff follows the original study [2]; the labels describe study groups rather than universal diagnostic categories.

Data source and preparation

The supplied GSE85307 files represent 157 participants in a nested case-control gene-expression study within VDAART: 110 normal pregnancies and 47 participants who subsequently developed preeclampsia [2,3]. These are whole-blood microarray expression measurements, not DNA genotypes. GEO documents RMA normalization on a log2 scale [4]. The supplied expression values are analyzed on their existing scale without another log transformation. The class file is a 500-probe subset, not the full transcriptome.

All preparation below occurs in R directly from unchanged copies of the supplied CSVs. The expression columns and covariate rows must be matched by sample identifier, not position.

```
covariates <- read.csv("data/GSE85307_covar-1.csv", stringsAsFactors = FALSE,
                       check.names = FALSE)
expression <- read.csv("data/GSE85307_geneExpression.csv", row.names = 1,
                       check.names = FALSE)
gene_names <- read.csv("data/GSE85307_geneNames.csv", stringsAsFactors = FALSE)
```

```r
# Stop rather than silently duplicating or losing participants.
stopifnot(!anyDuplicated(covariates$geoID),
          !anyDuplicated(covariates$subjectID),
          !anyDuplicated(colnames(expression)),
          !anyDuplicated(rownames(expression)),
          setequal(covariates$geoID, colnames(expression)))
original_order_matches <- identical(covariates$geoID, colnames(expression))
expression <- expression[, match(covariates$geoID, colnames(expression)), drop = FALSE]
stopifnot(identical(colnames(expression), covariates$geoID))

# Use the provided linker, retaining probe identifiers as identifiers.
cxcr2_probe <- as.character(gene_names$PROBEID[gene_names$GENE == "CXCR2"])
stopifnot(length(cxcr2_probe) == 1L, cxcr2_probe %in% rownames(expression))

# Values are prefixed by descriptive strings such as '...: 22.4'.
strip_prefix <- function(x) trimws(sub("^[^:]*:", "", x))
vitamin_d <- as.numeric(strip_prefix(covariates$mat_vitD))
outcome <- strip_prefix(covariates$preg_preeclampsia)
stopifnot(all(is.finite(vitamin_d)),
          all(outcome %in% c("Normal", "Preeclampsia")))

analysis <- data.frame(
  geoID = covariates$geoID,
  CXCR2 = as.numeric(expression[cxcr2_probe, ]),
  vitamin_d = vitamin_d,
  outcome = factor(outcome, levels = c("Normal", "Preeclampsia"),
                   labels = c("Normal pregnancy", "Preeclampsia")),
  vitamin_d_group = factor(ifelse(vitamin_d < 30, "<30 ng/mL", " 30 ng/mL"),
                           levels = c("<30 ng/mL", " 30 ng/mL"))
)
stopifnot(all(complete.cases(analysis)), all(is.finite(analysis$CXCR2)))
knitr::kable(data.frame(
  Check = c("Initial participants", "Final participants", "Excluded participants",
            "Expression probes", "CXCR2 probe ID", "Original sample order matched"),
  Result = c(nrow(covariates), nrow(analysis), 0, nrow(expression),
             cxcr2_probe, original_order_matches)
), caption = "Data alignment and completeness checks.")
```

Table 1: Data alignment and completeness checks.

Check	Result
Initial participants	157
Final participants	157
Excluded participants	0
Expression probes	500
CXCR2 probe ID	8048227
Original sample order matched	FALSE

All 157 participants are retained. There are no missing values in the selected analysis variables. Values that appear unusual are retained because no documented measurement error has been established.

```r
group_counts <- as.data.frame(table(analysis$outcome, analysis$vitamin_d_group))
names(group_counts) <- c("Pregnancy outcome", "Vitamin D category", "Participants")
knitr::kable(group_counts, caption = "Numbers of participants in each plotted subgroup.")
```

Table 2: Numbers of participants in each plotted subgroup.

Pregnancy outcome	Vitamin D category	Participants
Normal pregnancy	<30 ng/mL	78
Preeclampsia	<30 ng/mL	41
Normal pregnancy	30 ng/mL	32
Preeclampsia	30 ng/mL	6

Plot styling

All three figures are made with ggplot2. The same categorical palette is used throughout; each figure is also saved as a PDF and a 300-dpi PNG. Captions distinguish descriptive patterns from causal conclusions.

```r
vitamin_colors <- c("<30 ng/mL" = "#0072B2", " 30 ng/mL" = "#D55E00")
project_theme <- theme_classic(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 15),
        plot.subtitle = element_text(size = 11, color = "grey30"),
        legend.position = "bottom", plot.margin = margin(12, 16, 10, 10))
export_plot <- function(plot, name) {
  ggsave(file.path("figures", paste0(name, ".pdf")), plot,
         width = 8, height = 5.3, device = grDevices::cairo_pdf)
  ggsave(file.path("figures", paste0(name, ".png")), plot,
         width = 8, height = 5.3, dpi = 300)
}
```

Figure 1 · Expression distribution

```r
histogram <- ggplot(analysis, aes(x = CXCR2)) +
  geom_histogram(binwidth = 0.10, boundary = 0, fill = "#0072B2",
                 color = "white", linewidth = 0.4) +
  labs(title = "Distribution of CXCR2 expression",
       subtitle = "Maternal whole blood · 157 participants",
       x = "CXCR2 expression (log2 RMA signal)", y = "Number of participants") +
  project_theme
export_plot(histogram, "01_cxcr2_histogram")
knitr::include_graphics("figures/01_cxcr2_histogram.png")
```

CXCR2 expression ranges from 11.35 to 12.53 log2 units, with median 12.20 (Figure 1). The plot describes variation within this gene; it does not establish differences between clinical groups.

Figure 2 · Vitamin D and expression

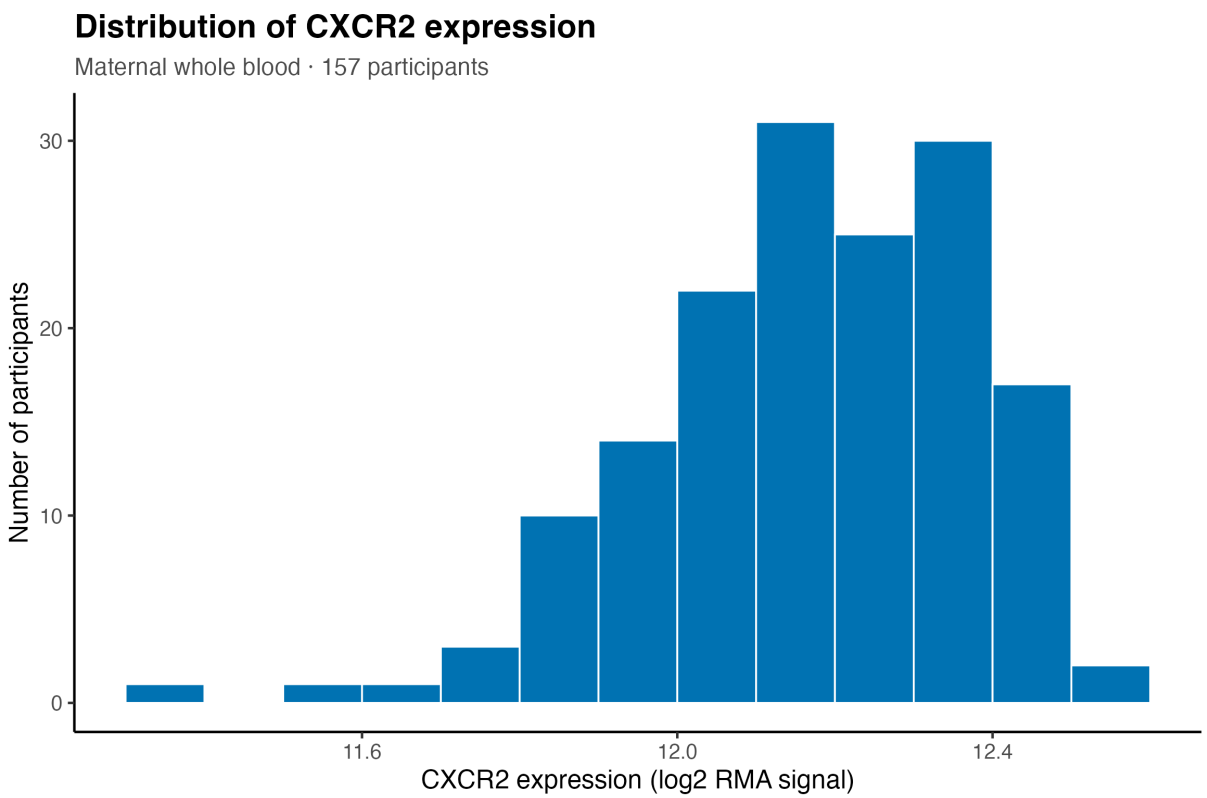


Figure 1: Figure 1. Distribution of CXCR2 expression across 157 participants. Histogram bins have a width of 0.10 log2 expression units. No observations were excluded.

```
scatterplot <- ggplot(analysis, aes(x = vitamin_d, y = CXCR2)) +
  geom_point(color = "#0072B2", size = 2, alpha = 0.65) +
  geom_smooth(method = "lm", formula = y ~ x, color = "#222222",
              fill = "#9DA9B5", linewidth = 0.8) +
  labs(title = "CXCR2 expression and baseline vitamin D",
       subtitle = "Exploratory, unadjusted relationship",
       x = "Baseline 25-hydroxyvitamin D (ng/mL)",
       y = "CXCR2 expression (log2 RMA signal)") + project_theme
export_plot(scatterplot, "02_cxcr2_vitamin_d")
knitr::include_graphics("figures/02_cxcr2_vitamin_d.png")
```

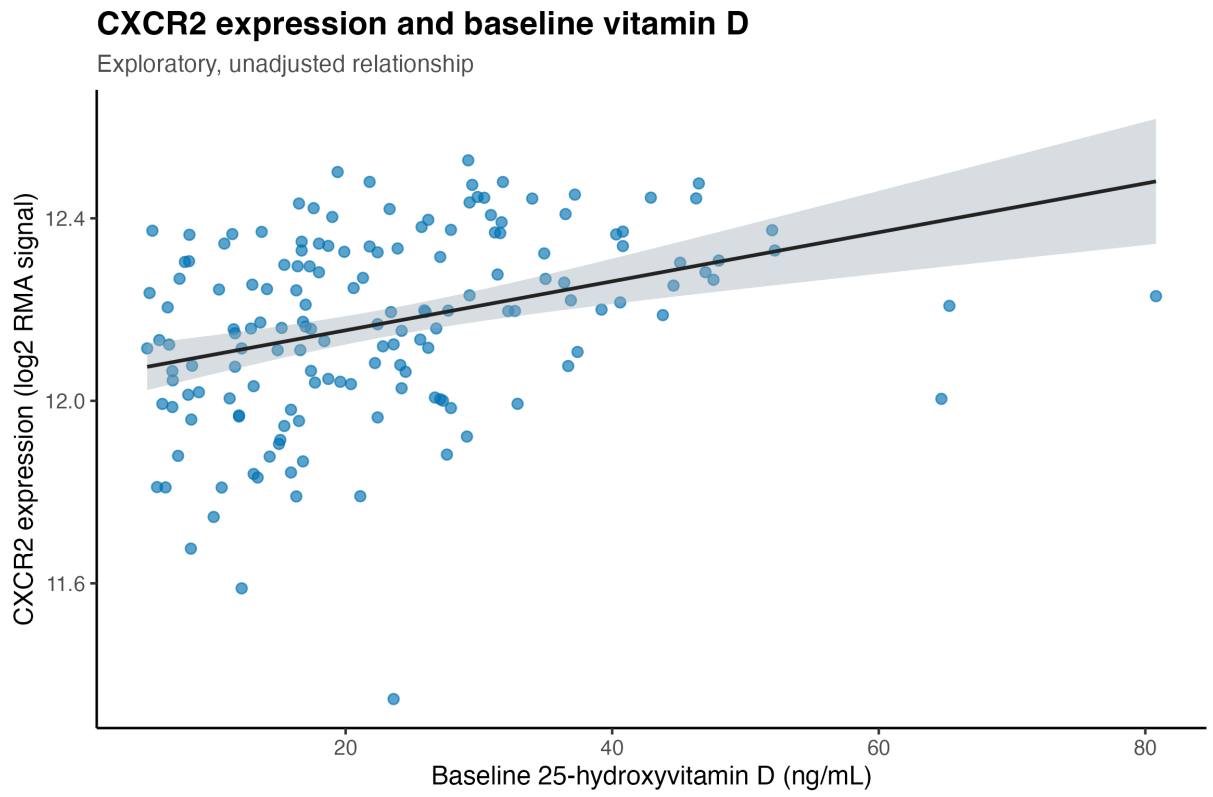


Figure 2: Figure 2. Baseline vitamin D concentration and CXCR2 expression. Each point represents one participant. The line is an unadjusted ordinary least-squares fit; the shaded band is its 95% confidence interval for the mean response, not a prediction interval.

The sample Pearson correlation is 0.35, indicating a positive unadjusted linear relationship (Figure 2). This descriptive relationship does not show that vitamin D changes CXCR2 expression. Batch, cell composition, and other participant characteristics could contribute.

Figure 3 · Expression by both categorical variables

```
counts <- aggregate(CXCR2 ~ outcome + vitamin_d_group, analysis, length)
counts$label <- paste0("n = ", counts$CXCR2)
```

```
counts$label_y <- max(analysis$CXCR2) + 0.10
boxplot <- ggplot(analysis, aes(x = outcome, y = CXCR2, fill = vitamin_d_group)) +
  geom_boxplot(position = position_dodge(width = 0.75), width = 0.6,
               outlier.shape = NA, alpha = 0.65, linewidth = 0.5) +
  geom_point(aes(group = vitamin_d_group), shape = 21, size = 1.6,
             alpha = 0.65, stroke = 0.2,
             position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.75, seed = 103)) +
  geom_text(data = counts, aes(y = label_y, label = label, group = vitamin_d_group),
            position = position_dodge(width = 0.75), size = 3.5) +
  scale_fill_manual(values = vitamin_colors, drop = FALSE) +
  scale_y_continuous(expand = expansion(mult = c(0.04, 0.08))) +
  labs(title = "CXCR2 expression across pregnancy and vitamin D groups",
       subtitle = "All observations shown; subgroup sizes are unequal",
       x = NULL, y = "CXCR2 expression (log2 RMA signal)",
       fill = "Baseline vitamin D") + project_theme
export_plot(boxplot, "03_cxcr2_grouped_boxplot")
knitr::include_graphics("figures/03_cxcr2_grouped_boxplot.png")
```

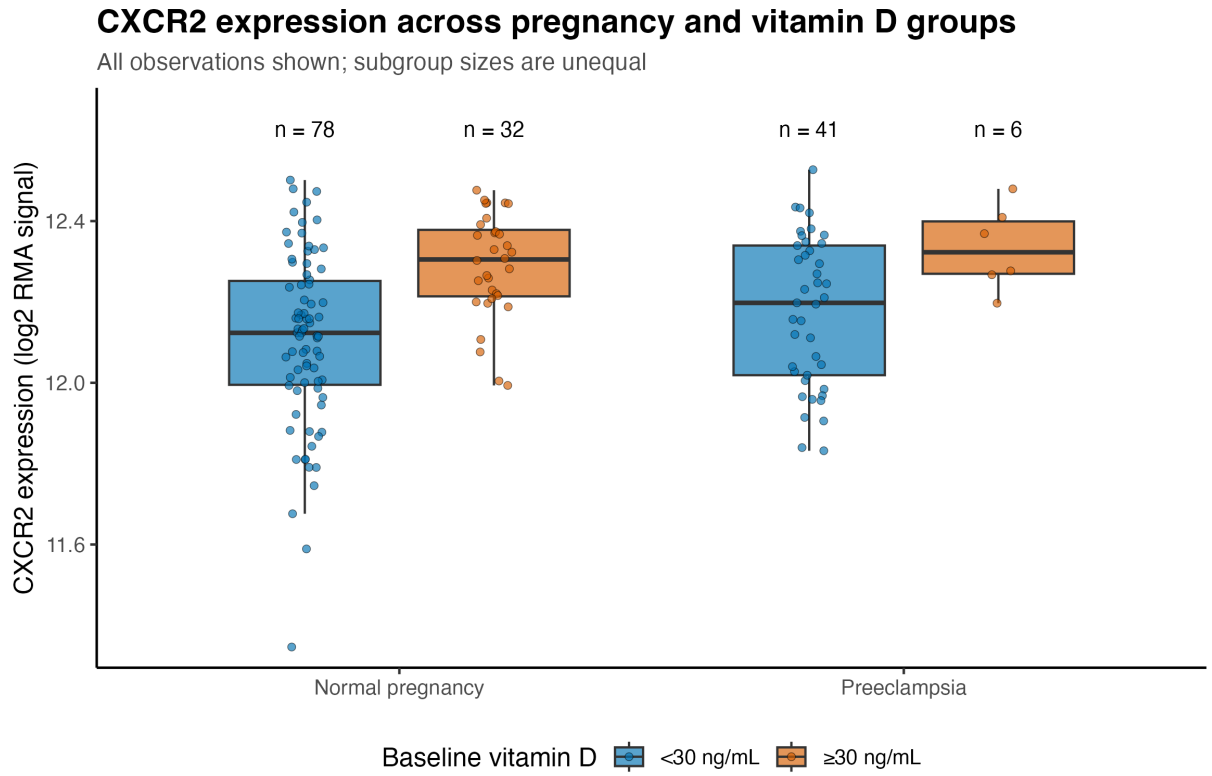


Figure 3: Figure 3. CXCR2 expression by pregnancy outcome and baseline vitamin D category. Boxes show the interquartile range, with a median line; whiskers extend to the most extreme observations within 1.5 interquartile ranges. All individual observations are overlaid with horizontal jitter. Subgroup counts appear above the boxes. The preeclampsia/ 30 ng/mL group contains only six participants.

```
medians <- aggregate(CXCR2 ~ outcome + vitamin_d_group, analysis, median)
knitr::kable(medians, digits = 3,
             col.names = c("Pregnancy outcome", "Vitamin D category", "Median CXCR2"),
             caption = "Descriptive subgroup medians supporting interpretation of Figure 3.")
```

Table 3: Descriptive subgroup medians supporting interpretation of Figure 3.

Pregnancy outcome	Vitamin D category	Median CXCR2
Normal pregnancy	<30 ng/mL	12.124
Preeclampsia	<30 ng/mL	12.198
Normal pregnancy	30 ng/mL	12.305
Preeclampsia	30 ng/mL	12.323

Group distributions overlap (Figure 3). Differences in medians should be interpreted descriptively, particularly for the six participants with preeclampsia and vitamin D 30 ng/mL. These plots do not demonstrate independent effects or an interaction between pregnancy outcome and vitamin D.

Interpretation and next steps

This first check-in describes one gene and the three required covariates. The sample is a selected pregnancy cohort, so its case fraction is not a population estimate of preeclampsia risk. Expression differences can also reflect blood-cell composition. The eight recorded microarray batches should be examined in any expanded analysis; the figures here are not batch-adjusted. No causal, diagnostic, or predictive claim is made.

For later submissions, CYP24A1 and PTX3 are biologically motivated additional genes available in the supplied data. An exploratory ten-gene clustering analysis could introduce unsupervised machine learning. Clinical labels would be withheld from the clustering inputs. This extension is planned, not performed in the first check-in.

References and software

1. NCBI Gene. *CXCR2, C-X-C motif chemokine receptor 2 (Homo sapiens).* https://www.ncbi.nlm.nih.gov/gene/3579
2. Mirzakhani H, et al. Early pregnancy vitamin D status and risk of preeclampsia. *Journal of Clinical Investigation.* 2016;126(12):4702–4715. https://doi.org/10.1172/JCI89031
3. Gene Expression Omnibus. GSE85307. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE85307
4. Gene Expression Omnibus. GSM2264465, data-processing description. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM2264465

The analysis and plots use R and ggplot2; knitr and rmarkdown generate this report. Exact citations and installed versions follow.

```
for (pkg in c("base", "ggplot2", "knitr", "rmarkdown")) {
  cat(paste(format(citation(pkg), style = "text"), collapse = "\n"), "\n\n")
}
```


R Core Team (2025). *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Wickham H (2016). *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. ISBN 978-3-319-24277-4, https://ggplot2.tidyverse.org.

Xie Y (2025). *knitr: A General-Purpose Package for Dynamic Report Generation in R*. R package version 1.50, https://yihui.org/knitr/. Xie Y (2015). *Dynamic Documents with R and knitr*, 2nd edition. Chapman and Hall/CRC, Boca Raton, Florida. ISBN 978-1498716963, https://yihui.org/knitr/. Xie Y (2014). "knitr: A Comprehensive Tool for Reproducible Research in R." In Stodden V, Leisch F, Peng RD (eds.), *Implementing Reproducible Computational Research*. Chapman and Hall/CRC. ISBN 978-1466561595.

Allaire J, Xie Y, Dervieux C, McPherson J, Luraschi J, Ushey K, Atkins A, Wickham H, Cheng J, Chang W, Iannone R (2024). *rmarkdown: Dynamic Documents for R*. R package version 2.29, https://github.com/rstudio/rmarkdown. Xie Y, Allaire J, Grolemund G (2018). *R Markdown: The Definitive Guide*. Chapman and Hall/CRC, Boca Raton, Florida. ISBN 9781138359338, https://bookdown.org/yihui/rmarkdown. Xie Y, Dervieux C, Riederer E (2020). *R Markdown Cookbook*. Chapman and Hall/CRC, Boca Raton, Florida. ISBN 9780367563837, https://bookdown.org/yihui/rmarkdown-cookbook.

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.7.3
##
## Matrix products: default
## BLAS:
  /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK:
  /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
  LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] ggplot2_3.5.2
##
## loaded via a namespace (and not attached):
## [1] vctrs_0.7.3 nlme_3.1-168 cli_3.6.5 knitr_1.50
## [5] rlang_1.3.0 xfun_0.52 generics_0.1.4 textshaping_1.0.1
## [9] labeling_0.4.3 glue_1.8.0 htmltools_0.5.8.1 ragg_1.4.0
## [13] scales_1.4.0 rmarkdown_2.29 grid_4.5.1 evaluate_1.0.4
## [17] tibble_3.3.0 fastmap_1.2.0 yaml_2.3.10 lifecycle_1.0.5
## [21] compiler_4.5.1 dplyr_1.2.1 RColorBrewer_1.1-3 pkgconfig_2.0.3
## [25] mgcv_1.9-3 rstudioapi_0.17.1 lattice_0.22-7 systemfonts_1.2.3
## [29] farver_2.1.2 digest_0.6.37 R6_2.6.1 tidyselect_1.2.1
## [33] splines_4.5.1 pillar_1.10.2 magrittr_2.0.3 Matrix_1.7-3
## [37] withr_3.0.2        tools_4.5.1        gtable_0.3.6
```